CELLFEE · QpiAI outputs (recorded runs)

Pipeline run: 2026-10-30 02:09:46 · SDK demo: 2026-10-30 02:43:56 · QpiAI Quantum SDK 0.3.0 · QCloud key configured: True · QpiAI Opt key configured: True · devices: QpiAI-QSV-Local
Everything below is read from artifacts/results.json, artifacts/sdk_report.json and artifacts/evidence/.
Reproduce: cd backend; python -m cellfee.sdk demo (about 5 min) or python -m cellfee.pipeline --retrain (about 70 min).

WHAT QPIAI COMPUTES

```
    QpiAI QCloud   feature-map circuits -> quantum kernel (similarity of two batteries); shot-based
  fidelity circuits; QAOA
    QpiAI Opt      every AQSVC SVM trained as an 80-variable QUBO (600 QUBOs this run);
                   fleet reuse/test/recycle + packs for 5,000 batteries in 152 batches (exact MILP); market
  matching
```

VALIDATED RESULT (grouped 5-fold cross-validation by cell, 8,753 real snapshots, 124 LFP cells)

```
    AQSVC (quantum: QpiAI kernel + QpiAI Opt QUBOs)   accuracy 95.06%   AUC 0.990
    XGBoost (classical)                               accuracy 94.95%   AUC 0.992
    Both together (CELLFEE grade)                     accuracy 95.64%   AUC 0.993
    Same quantum ensemble, classical solver (C1)      accuracy 94.87%
    Models agree on 95.54% of snapshots and are right 97.11% of those.
    AQSVC ensemble: 25 bagged + 25 boosted QUBO-SVMs on each of 2 tuned feature maps, stacked (plain vote
  averages 92.32% over the folds).
```

FLEET DECISIONS (QpiAI Opt)

```
    152 of 152 batches solved exactly by QpiAI Opt's MILP; 10 of 10 re-solved by HiGHS give the same
  objective; 452 packs; 0 forbidden assignments; all feasible: True.
```

EVIDENCE: 174 run records in artifacts/evidence/ · fleet_allocation 42, allocation 30, market_matching 19, quantum_kernel_fidelity_test 16, qaoa_pack_balance 13, twin_verification 12, aqsvc_training 10, assessment 7, usage_baseline_models 5, upload_screening 5

No quantum advantage is claimed: circuits run on QpiAI's local statevector simulator (classical simulation).

CELLFEE SDK · live QpiAI runs and reproducibility

python -m cellfee.sdk demo: each step is a live QpiAI run with an evidence record (run id in brackets).

kernel [1030-024402-24e1] QpiAI SDK statevector overlap 0.0028 · 2000-shot fidelity circuit on QpiAI-QSV-Local: 0.0050 · NumPy twin difference 1.7e-16
train [1030-024410-4e07] QpiAI Opt local QUBO annealer: 80 binary variables, 4 reads x 366 sweeps, 3.92 s · energy -6.765 (built-in annealer -9.003) · held-out accuracy 83.60% (exact classical solution 86.00%)
screen [1030-024413-7653] XGBoost 0.239 · AQSVC 0.122 (46 of 100 QUBO-SVMs say fit) · grade C -> recycle · circuit 7 qubits, depth 18
allocate [1030-024508-b0ee] batch 1, 33 batteries, solved by QpiAI Opt local MILP (exact), objective ₹515.818, feasible True

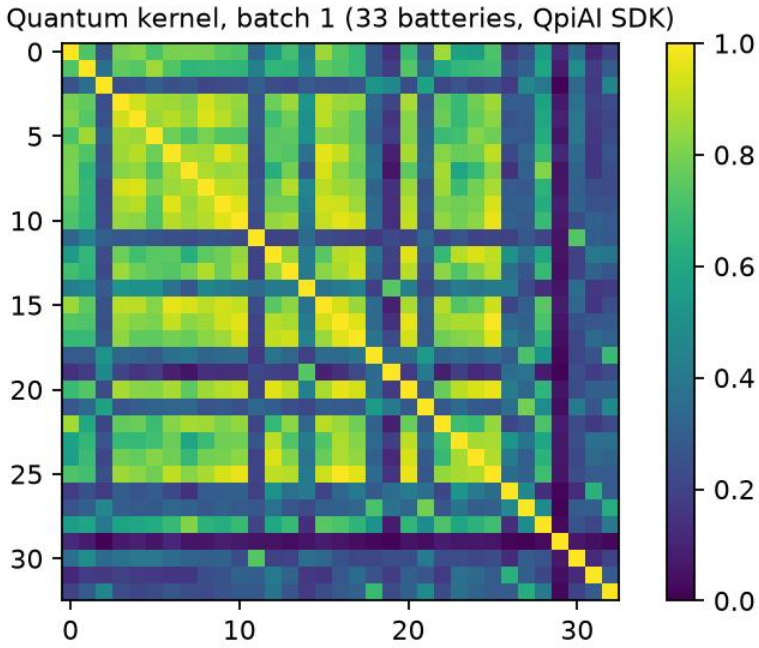
```
    Exact MILP (HiGHS)                                objective 515.8182  time 25.99 s
    QpiAI Opt local MILP (exact)                      objective 515.8182  time 24.89 s
    QpiAI Opt local QUBO annealer (free-tier budget) objective 200.9335  time 2.92 s
```

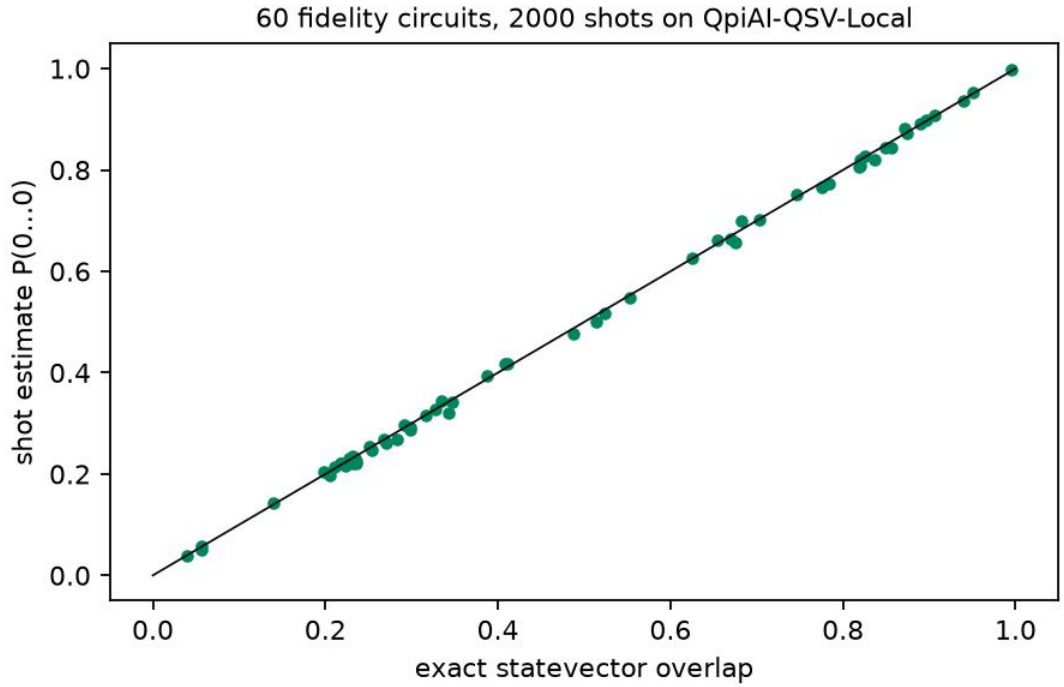
qaoa [1030-024512-15fb] XY-mixer QAOA (p=3, CVaR α=0.25) on QpiAI QCloud circuits · 8 qubits on QpiAI-QSV-Local · P(optimal split) 0.060 vs random 0.029 · approximation ratio 1.000
match [1030-024513-ed08] QpiAI Opt local solver: 7 buyer-pack matches from 42 compatible pairs, total score 4.6424

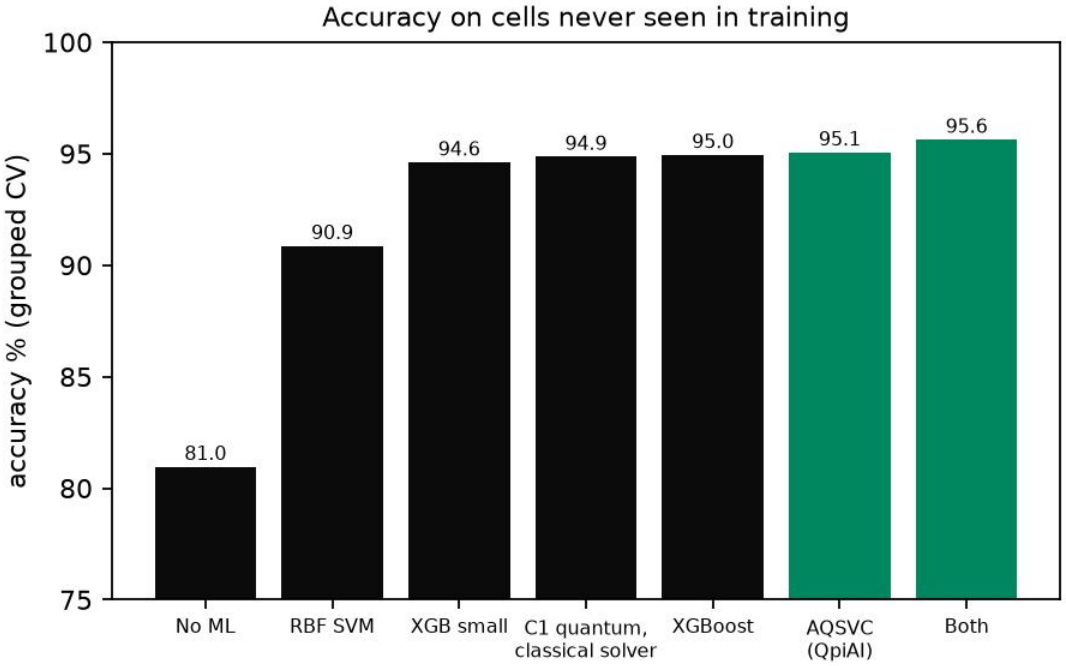
verify [1030-024604-dbca] 7 / 7 reproducibility checks passed

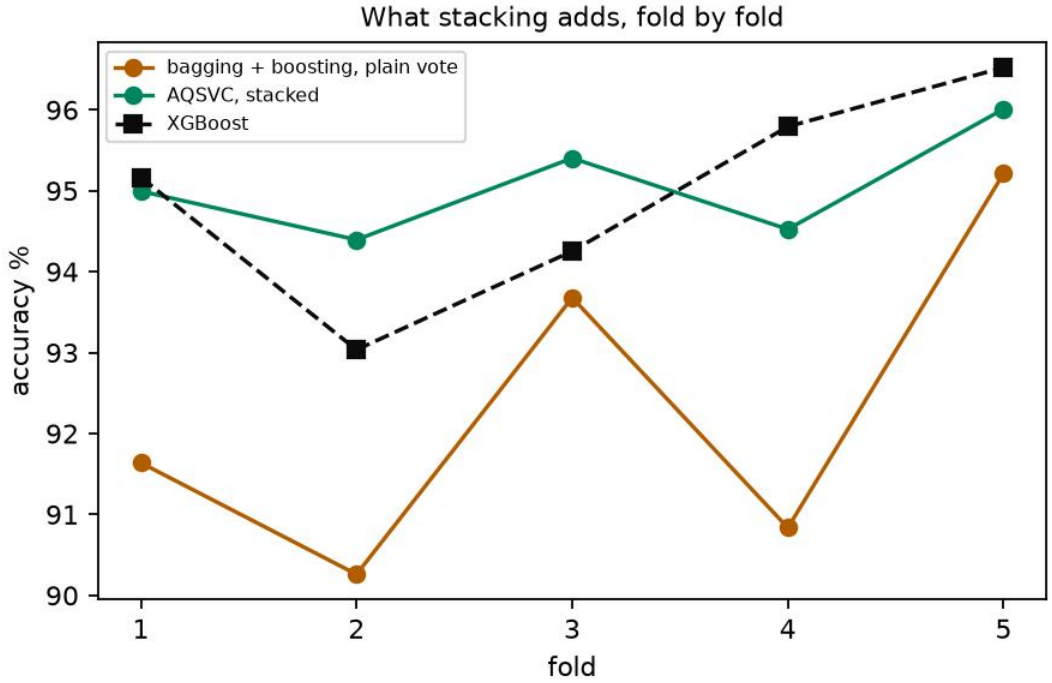
```
    PASS  QpiAI SDK kernel (batch 1) reproduces the recorded kernel — max |difference| 2.34e-04 (recorded
  to 4 decimals, angles to 4 decimals)
    PASS  NumPy training twin equals the QpiAI SDK circuit — max |difference| 2.66e-15
    PASS  Recorded shot-based fidelity circuits agree with exact overlaps — mean |shots - exact| 0.0066
  over 60 circuits, 2000 shots
    PASS  QpiAI Opt re-solves batch 1 to the recorded objective — now 515.8182 vs recorded 515.8223 (solved
  by QpiAI Opt local MILP (exact))
    PASS  QpiAI Opt exact MILP matches the HiGHS reference — QpiAI Opt 515.818164 vs HiGHS 515.818164
    PASS  QpiAI Opt market matching reproduces the recorded total score — now 4.6424 vs recorded 4.6424
    PASS  Evidence log present — 165 run records in artifacts/evidence/
```

CELLFEE · QpiAI outputs at a glance









Evidence records

One recent record of each kind, abridged (full JSON in artifacts/evidence/<run_id>.json and on the dashboard's Evidence log).

sdk_market_matching · run 1030-025107-e8d9 · 2026-10-30T02:51:07 · backend: QpiAI Opt · device: —

```
SDK match: 7 buyer-listing matches, total score 4.642
```

sdk_verify · run 1030-025107-98b7 · 2026-10-30T02:51:07 · backend: QpiAI QCloud SDK + QpiAI Opt · device: —

```
SDK verify: 7/7 checks passed
```

market_matching · run 1030-025107-2581 · 2026-10-30T02:51:07 · backend: QpiAI Opt local solver · device: —

```
7 matches from 42 compatible pairs
```

allocation · run 1030-025106-36d8 · 2026-10-30T02:51:06 · backend: QpiAI Opt modeller + QpiAI Opt solver · device: —

```
33 batteries, 3 packs, solved by QpiAI Opt local MILP (exact), objective 515.818
```

sdk_screen · run 1030-024749-6f28 · 2026-10-30T02:47:49 · backend: QpiAI QCloud SDK (circuits) + QUBO-trained AQSVC · device: QpiAI-QSV-Local

```
SDK screen: XGBoost 0.24, AQSVC 0.12, grade C
```

aqsvc_training · run 1030-024728-08db · 2026-10-30T02:47:28 · backend: QpiAI Opt (QUBO) + QpiAI QCloud feature-map kernel · device: —

```
1 AQSVC members trained as QUBOs (QpiAI Opt local QUBO annealer), 80 binary variables each
```

quantum_kernel_fidelity_test · run 1030-024710-ff40 · 2026-10-30T02:47:10 · backend: QpiAI QCloud SDK · device: QpiAI-QSV-Local

```
1 fidelity-test circuits on QpiAI-QSV-Local
```

qaoa_pack_balance · run 1030-024512-15fb · 2026-10-30T02:45:12 · backend: QpiAI QCloud SDK circuits + QpiAI Opt QAOA · device: QpiAI-QSV-Local

```
8-qubit QAOA: P(optimal)=0.060 vs random 0.029; exact objective 1.203
```

marketplace_listing · run 1030-022255-c651 · 2026-10-30T02:22:55 · backend: CELLFEE marketplace checks · device: —

```
listing L-261030-BB7FF6 submitted: Report uploaded — not yet verified
```

assessment · run 1030-021552-7221 · 2026-10-30T02:15:52 · backend: XGBoost + AQSVC (deployed) + usage-baseline models · device: —

```
live assessment: grade A, Safe, risk Low
```

fleet_allocation · run 1030-020946-f771 · 2026-10-30T02:09:46 · backend: HiGHS exact MILP (comparison variant) · device: —

```
xgb_small: 20 batches, value ₹10489.55, 0 forbidden assignments
```

twin_verification · run 1030-014117-8c6f · 2026-10-30T01:41:17 · backend: QpiAI QCloud SDK · device: QpiAI-QSV-Local

```
NumPy twin vs QpiAI SDK kernel: max diff 1.55e-15
```